
ShadowWeave: Calibrated Amodal Occupancy Completion into Occluded Space from a Single Depth Frame

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Abstract

A mobile agent must reason about space it cannot currently see — the region occluded by the walls and obstacles it is navigating around (“shadow”). We study how well a model can **complete** occupancy in that occluded region from a **single monocular-depth frame**, and, crucially, whether it knows *when it is guessing*. ShadowWeave projects one depth frame into an egocentric bird’s-eye-view (BEV) occupancy grid, marks the unobserved cells with an explicit shadow mask, and predicts occupancy inside that mask together with a per-cell uncertainty. On a MuJoCo indoor benchmark, the model beats a persistence baseline by a micro-averaged shadow-IoU gain of **+0.31 to +0.36 at a 5 s horizon** (95% per-episode bootstrap CIs clear of zero at every horizon), far exceeds an observation-blind dataset prior even at the prior’s best threshold (0.61–0.72 IoU vs 0.08–0.10), and its uncertainty is **better calibrated inside shadow (ECE 0.043–0.047) than in directly observed space (0.058–0.086)**. We introduce a decomposition that separates *amodal completion* of persistent hidden structure from *temporal forecasting* of dynamics, and use it to make an honest claim: the gain is almost entirely completion — the pure-forecasting residual sits at the noise floor. Randomizing room geometry per episode leaves the gain statistically unchanged on rooms the model never saw, ruling out memorization of a fixed layout. The completed grid and its uncertainty drive an eyes-free spatial-audio navigation interface at 7 ms (p95) perception-to-planning latency, well inside a 20 Hz budget. We release the evaluation methodology — an empty-credit-immune micro-averaged gain, a completion/forecasting decomposition, and per-episode confidence intervals — as a reusable protocol for honest occupancy-completion claims.

1 Introduction

Agents act in space they cannot see. A person stepping into a corridor, a robot rounding a shelf, a blind traveler crossing an unfamiliar room — each must anticipate structure that is occluded by the very obstacles they are navigating around. Sensing provides a partial view; the rest must be inferred. For safety-critical, eyes-free use, that inference is only useful if it is accompanied by an honest estimate of its own reliability: a confident hallucination of clear space behind a wall is worse than a calibrated “I don’t know.”

Prior work addresses parts of this problem but not their conjunction. Semantic scene completion, from MonoScene [Cao and de Charette, 2022] through VoxFormer [Li et al., 2023] and VisHall3D

[Lu et al., 2025], hallucinates dense geometry beyond the visible frontier from a single image, but targets outdoor driving voxels scored by mIoU alone, with no notion of calibrated confidence in the hallucinated region. Occupancy world models such as OccWorld [Zheng et al., 2024] forecast the *temporal* evolution of multi-frame occupancy for autonomous driving. Closest to our setting, ProxMaP [Sharma et al., 2023] completes indoor proximal occupancy from a single top-down view to improve navigation — but it emits a point estimate, with no explicit occlusion mask, no calibrated uncertainty in the hidden cells, and no attribution of *where* its predictive power comes from.

We present **ShadowWeave**, which from a single monocular-depth frame produces (i) an egocentric BEV occupancy grid, (ii) an explicit shadow mask marking the cells the sensor cannot see, and (iii) a calibrated per-cell occupancy probability inside that mask. We evaluate it with a protocol designed to resist the standard failure modes of occupancy metrics: an empty-credit-immune, micro-averaged shadow-IoU gain over a persistence baseline; per-episode bootstrap confidence intervals; and a static/dynamic/never decomposition that attributes the gain to *completion* of persistent hidden structure versus *forecasting* of dynamics. Downstream, the completed grid and its uncertainty drive an A* planner and a nine-zone HRTF spatial-audio interface for eyes-free navigation.

Our contributions are:

- **A calibrated amodal-completion result.** From one depth frame, the model completes occluded occupancy with a micro-averaged shadow gain of +0.31–0.36 at 5 s (CIs clear of zero at every horizon), and its in-shadow uncertainty is *better* calibrated (ECE 0.043–0.047) than in observed space — evidence that the model conveys, rather than merely produces, hidden-space predictions.
- **A decomposition methodology** that separates completion from forecasting. It reveals that the gain is spatial completion of persistent hidden structure; the pure-forecasting residual is at the noise floor (fixed static +0.000; moving tier +0.007 [+0.002, +0.012] at 5 s — statistically significant but practically small, and decaying with horizon as real forecasting should). We report this negative honestly: most pooled occupancy numbers in the literature are vulnerable to exactly the confound this decomposition dissects.
- **A memorization confound-kill.** Randomizing room geometry per episode leaves the gain statistically unchanged on never-seen rooms (+0.306 fixed vs +0.309 [+0.294, +0.324] randomized), showing the completion is per-scene inference, not a memorized constant layout.
- **A real-time eyes-free system** that turns completed occupancy and its uncertainty into an A* path and nine-zone spatial-audio cues at 7 ms p95 perception-to-planning latency.

2 Method

2.1 From a depth frame to an egocentric BEV with a shadow mask

At each step the agent observes a single monocular-depth image (in simulation, the renderer’s depth; in deployment, a monocular-depth network such as Depth-Anything-V2-Small). A differentiable projector lifts the depth map into an egocentric bird’s-eye-view (BEV) occupancy grid of size 96×96 covering 5 m ahead, together with a **visibility** grid: cells along a ray up to the first returned surface are marked observed-free, the surface cell observed-occupied, and everything beyond the surface **shadow** — geometry the sensor cannot see. Thresholding visibility at 0.5 yields the binary shadow mask that defines the region of interest for every metric in this paper (Fig. 1). Optical flow between consecutive BEV frames provides an optional motion channel.

2.2 World model

A U-Net (31M parameters) takes the single-frame BEV stack and predicts occupancy at horizons of 1, 3, 5, and 10 s as per-cell probabilities. It is trained with binary cross-entropy (positive class up-weighted, since BEV targets are $\approx 7\%$ occupied) on 400 training episodes, with early stopping

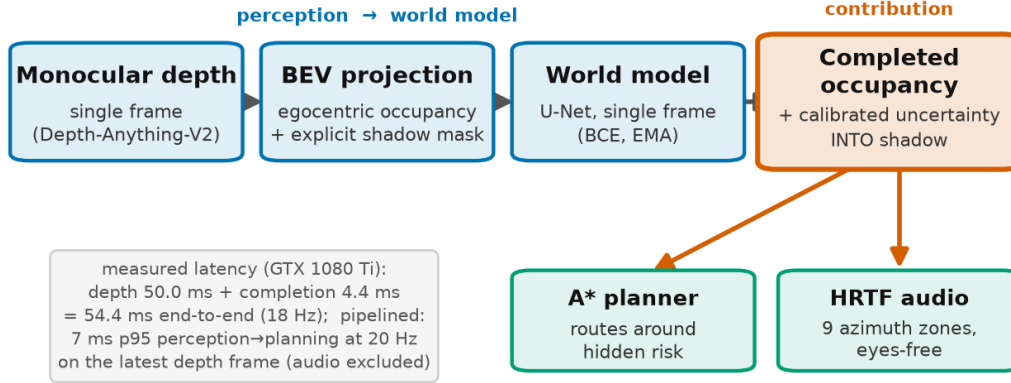


Figure 1: ShadowWeave pipeline. A single monocular-depth frame is lifted by a differentiable projector into an egocentric 96×96 BEV occupancy grid together with a visibility grid; thresholding visibility yields the explicit shadow mask marking cells the sensor cannot see. A 31M-parameter U-Net predicts per-cell occupancy probabilities at 1/3/5/10 s horizons inside that mask; the completed grid and its uncertainty feed an A* planner and a nine-zone HRTF spatial-audio interface for eyes-free navigation.

and exponentially-averaged (EMA) weights; we report the epoch-8 checkpoint (validation loss 0.300, validation IoU@5s 0.665). Ablations (§4.6) show the visibility and flow channels are redundant — the completion signal is carried by the occupancy channel alone — so the deployable model reduces to depth→occupancy→completion. The input is a **single frame**: the model has no multi-frame memory, which is what makes the completion-vs-forecasting distinction (§3.2) sharp.

2.3 Uncertainty in shadow

The model’s per-cell occupancy probability *is* its uncertainty estimate; the claim we test in §4.5 is that this probability is **calibrated inside the shadow region specifically**, not merely grid-wide (where easy free space dominates). Calibrating uncertainty in the occluded region is the property that makes the output safe to consume downstream.

2.4 Downstream eyes-free interface (system context)

The completed occupancy grid and its uncertainty drive an A* global planner (which owns goal-seeking and pays an extra cost to route through unobserved cells) and a reactive local collision-avoidance policy. Occupancy and uncertainty in nine azimuth zones are rendered as HRTF-spatialized audio cues for eyes-free navigation. We describe this interface for context and report its latency and navigation behavior honestly (§4.7); the audio interface itself is a system demonstration, not a user-evaluated result.

3 Evaluation methodology

Occupancy metrics are easy to report and easy to inflate. Because our claim lives entirely inside a sparse, mostly-empty sub-region (shadow is $\sim 92\%$ never-occupied), we adopt a protocol built to resist the specific ways such a claim can look better than it is. We consider this protocol a contribution in its own right.

102 3.1 The empty-credit trap and micro-averaging

103 The natural masked IoU scores an empty prediction against an empty masked region as a perfect 1.0
104 (“correctly predicted nothing here”). Since most shadow cells are legitimately empty, this inflates
105 both the model and the baselines, and the inflation is not uniform across sub-masks. We therefore
106 report **micro-averaged** shadow IoU: pool integer intersection and union counts across all frames
107 and divide once, so rare sub-masks cannot each contribute a free 1.0. Empty denominators report as
108 undefined, never as 1.0. The defended quantity is the **gain** over a persistence baseline (the observed
109 frame carried forward), not the absolute level.

110 3.2 Completion vs. forecasting decomposition

111 Within the shadow mask, we classify each cell by its occupancy pattern across the prediction horizons
112 — valid because all horizons are rendered from the same issue pose, so static geometry projects to
113 identical cells at every horizon:

- 114 • **STATIC** — occupied at *all* horizons: persistent hidden structure (walls, static obstacles,
115 settled debris); $\approx 7\%$ of shadow. Recall here measures **amodal completion**.
- 116 • **DYNAMIC** — occupied at *some but not all* horizons: transient-in-window structure (arrivals,
117 departures, pass-throughs); $\leq 1.5\%$ of shadow. IoU here measures genuine **forecasting**
118 beyond persistence.
- 119 • **NEVER** — occupied at *no* horizon: free hidden space; $\approx 92\%$ of shadow. The false-positive
120 test.

121 Reporting completion and forecasting separately is what lets us state honestly that the gain is
122 completion, not clairvoyance — a distinction the pooled shadow-IoU hides.

123 3.3 Per-episode bootstrap confidence intervals

124 Frames within an episode share geometry and pose and are correlated, so resampling frames would
125 understate uncertainty. We resample **episodes** with replacement ($B = 10,000$), recompute the pooled
126 micro-gain per replicate, and take the paired model-minus-persistence difference within each replicate
127 (the two are positively correlated across episodes). We report the resulting 2.5/97.5 percentile interval
128 for every headline gain.

129 3.4 Calibration in shadow

130 We measure expected calibration error (ECE) restricted to shadow cells, contrasted with observed cells
131 and with a walls-excluded variant, using equal-mass binning (a fine 1000-bin histogram merged into
132 15 equal-count macro-bins, because in-shadow probabilities cluster near the prior and equal-width
133 bins would be mostly empty). ECE-in-shadow close to or below ECE-in-observed is the evidence
134 that uncertainty propagates faithfully into unseen space.

135 4 Experiments

136 4.1 Setup

137 We use a MuJoCo single-room benchmark with three difficulty tiers — **static** (fixed obstacles),
138 **moving** (kinematically driven movers), and **debris** (objects that fall and settle) — with 400 training
139 and 80 validation episodes on disjoint seeds; the per-tier decomposition and CI analyses use clean
140 single-tier validation sets of 30 episodes (9,000 frames) each, and the randomized-geometry validation
141 (§4.4) uses 80 episodes. The BEV is 96×96 over 5 m; horizons are 1/3/5/10 s. Unless noted, fixed-
142 geometry results use a 6×6 m room; §4.4 randomizes geometry. Reproducibility is pinned: the
143 fixed-geometry scene generation is verified bit-exact against a golden hash, and a full rollout re-run

Table 1: Micro-averaged shadow-IoU gain over the persistence baseline at the 5 s horizon on the fixed-geometry validation set, per tier, with 95% per-episode bootstrap confidence intervals.

tier (fixed val, @5 s)	micro shadow gain	95% CI
static	+0.306	[+0.288, +0.327]
moving	+0.363	[+0.349, +0.378]
debris	+0.315	[+0.281, +0.350]

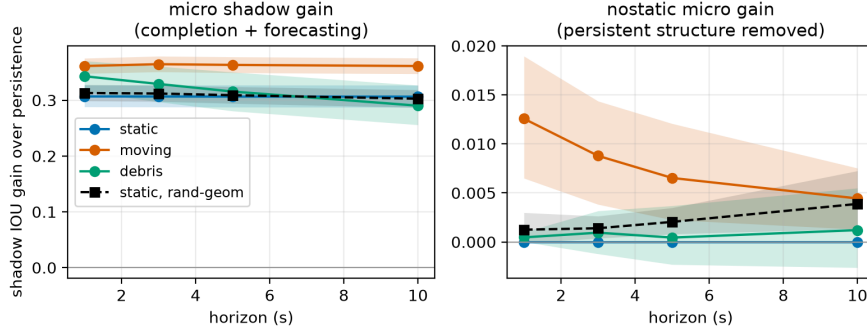


Figure 2: Micro-averaged shadow-IoU gain over persistence per tier and horizon, with 95% per-episode bootstrap confidence intervals. Every interval is clear of zero at every horizon, so the gain is statistically significant rather than sampling noise.

reproduces every accuracy metric bit-identically (wall-clock latency, which is hardware-dependent, is the one exception).

4.2 Shadow-gain headline

The model predicts occupancy in cells the sensor cannot see, beating persistence at every horizon. Micro-averaged shadow gain at 5 s, per tier, with 95% bootstrap CIs, is shown in Table 1.

Every interval is clear of zero at every horizon, so the gain is statistically significant rather than sampling noise (Fig. 2; all horizons tabulated in Appendix A). In policy rollouts the frame-averaged (macro) shadow gain is +0.325/+0.322/+0.320/+0.178 at 1/3/5/10 s. (Raw macro shadow-IoU levels — model 0.744 vs persistence 0.424 at 5 s — are empty-credit inflated and are reported only for reference; the micro-gain is the defended number.)

Observation-blind prior. Persistence could be a weak baseline for *completion*, so we also evaluate a pose-marginal dataset prior — the mean training-set occupancy in the egocentric frame, using no observation at all — swept over thresholds and reported at its most favorable one. It reaches only 0.08–0.10 micro shadow IoU across tiers, and 0.09 on the randomized-geometry set: to recall hidden structure (0.87–0.92) it must blanket 72–79% of genuinely empty hidden space with false positives, whereas the model achieves 0.61–0.72 IoU at a 1.4–3.4% false-positive rate. A complementary observation-conditioned heuristic — dilating every observed obstacle into the shadow, swept over radii and scored at its best — peaks at its smallest radius with micro shadow IoU 0.31–0.35, at or marginally below persistence itself; growing the radius only trades false positives for recall, monotonically lowering IoU. Hidden structure is non-local: neither copying the observation, extrapolating it locally, nor an observation-blind prior recovers it. Completion is therefore learned observation-conditional inference; together with the geometry-randomization result (§4.4) this rules out memorized layouts, memorized marginals, and trivial extrapolation alike (full tables in Appendix B).

Table 2: Completion/forecasting decomposition of the shadow mask on the fixed-geometry validation set at 5 s (dynamic IoU rows span the 1 s→10 s horizons). The gain is carried by recall of persistent hidden structure (amodal completion), not by forecasting of dynamics.

component (fixed val @5 s)	model	persistence
persistent-structure recall (amodal completion)	0.83–0.93	0.37–0.51
dynamic IoU, moving tier (1 s→10 s)	0.092 → 0.031	0.060 → 0.020
dynamic IoU, debris tier	0.020 → 0.019	0.006 → 0.023
false-positive rate on never-occupied cells	0.014–0.034	0.022–0.027

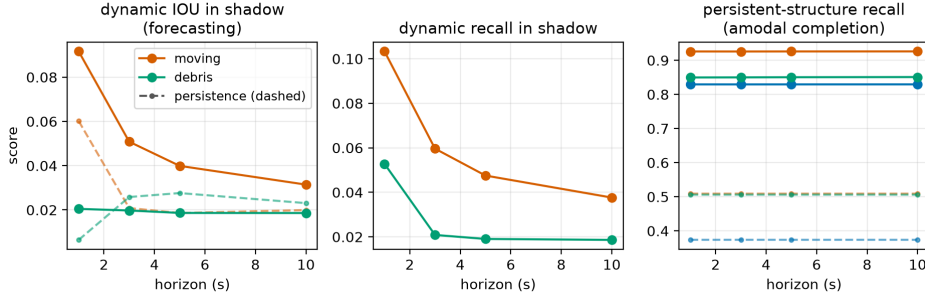


Figure 3: Decomposition of the shadow gain across horizons: persistent-structure recall (amodal completion) versus dynamic IoU (forecasting) for model and persistence. The gain is spatial completion of persistent hidden structure; the small dynamic-forecasting signal on the moving tier decays with horizon as genuine forecasting should.

167 4.3 What the gain is: completion, not clairvoyance

168 Decomposing the shadow mask (§3.2) attributes the gain almost entirely to amodal completion of
 169 persistent hidden structure (Table 2, Fig. 3).

170 Excluding all persistent structure, the residual “nostatic” gain sits at the noise floor: fixed static
 171 +0.000, moving +0.007 (CI [+0.002, +0.012]), randomized-geometry static +0.002 ([+0.001, +0.004])
 172 at 5 s. The honest claim is therefore **amodal completion of hidden structure**, with a small but
 173 statistically significant dynamic-forecasting signal on the moving tier that decays with horizon as
 174 genuine forecasting should. We do not claim dynamic forecasting on debris, where the dynamic
 175 signal is at the noise floor and persistence matches the model beyond 1 s.

176 4.4 Robustness to room geometry (memorization confound-kill)

177 A fixed room shell could let the model memorize a constant wall prior. We regenerated the data with
 178 per-episode randomized room width and depth (each drawn i.i.d. in [5, 8] m, obstacle counts scaled
 179 to hold the ~7% positive fraction), retrained the model from scratch, and evaluated on 80 held-out
 180 randomized rooms (Table 3).

181 The gain does not drop — the intervals overlap almost entirely, and completion recall is if anything
 182 slightly higher on never-seen rooms. Memorizing a constant shell contributes essentially nothing;
 183 completion is per-scene inference.

184 4.5 Calibration in shadow

185 At the 5 s horizon, in-shadow ECE is 0.043–0.047 across tiers — *better* than in observed space (0.058–
 186 0.086) — with Brier 0.022–0.036 (Fig. 4); excluding the persistent structure entirely, in-shadow
 187 ECE remains 0.055–0.056 on the tiers with dynamic content (on the static tier the structure-excluded
 188 region has no positive labels, so no calibration curve exists there), so the calibration is not carried
 189 by the easy always-occupied cells. The model is underconfident in the mid-range but places little

Table 3: Static-tier shadow gain at 5 s under fixed versus per-episode randomized room geometry. The gain does not drop on never-seen rooms, ruling out memorization of a constant layout.

static tier, @5 s	micro shadow gain	95% CI	completion recall
fixed 6×6 geometry	+0.306	[+0.288, +0.327]	0.829
randomized geometry	+0.309	[+0.294, +0.324]	0.854

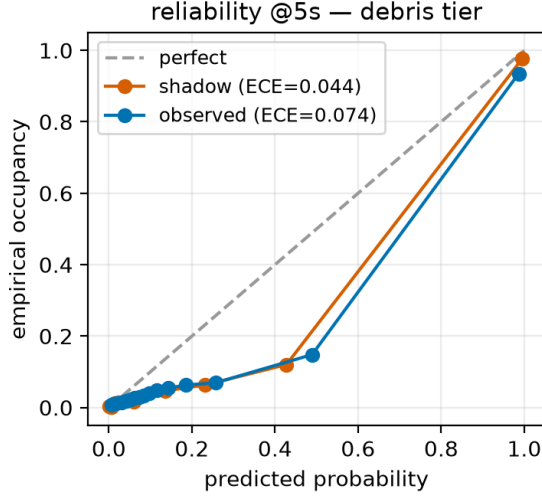


Figure 4: Reliability diagrams restricted to shadow cells (equal-mass binning: a fine 1000-bin histogram merged into 15 equal-count macro-bins). At 5 s, in-shadow ECE is 0.043–0.047 across tiers — better than in directly observed space (0.058–0.086) — with Brier 0.022–0.036; the model is underconfident in the mid-range but places little mass there.

190 mass there. Calibration is not an artifact of the fixed room: on randomized geometry, in-shadow ECE
 191 is 0.050–0.067 vs observed 0.056–0.077, with shadow becoming the better-calibrated region at the
 192 longer 5–10 s horizons.

193 4.6 Ablations: occupancy alone carries the signal

194 Dropping either auxiliary input channel and retraining leaves the gain intact (Table 4).

195 The no-visibility per-tier gains are within CI of the full model; the no-flow rollout gain is if anything
 196 higher. Both auxiliary channels are redundant, which simplifies the input and removes optical-flow
 197 estimation from the runtime critical path. These independently retrained variants also bound training-
 198 run variance: every one of them (and the separately trained randomized-geometry model of §4.4)
 199 lands within the full model’s per-tier confidence intervals, so the headline gain is not an artifact of a
 200 lucky training seed.

201 4.7 System: latency, anticipation, navigation

202 **Latency.** The perception-to-planning forward path (BEV projection, raycaster, world-model for-
 203 ward, orchestrator; batch 1, deterministic U-Net) runs at 5.97 ms p50 / 7.00 ms p95 — well inside a
 204 50 ms (20 Hz) budget. The monocular-depth stage is excluded and is not free: measured in isolation,
 205 Depth-Anything-V2-Small takes 49.9 ms p50 / 50.7 ms p95 on the evaluation-era GPU (GTX 1080 Ti,
 206 480×640 input) — comparable to the entire budget on that hardware. Run sequentially, the measured
 207 end-to-end latency including depth is 54.4 ms (18 Hz) on the same GPU; a deployed system therefore
 208 runs depth asynchronously at its own ~ 20 Hz rate while the 7 ms completion path consumes the most
 209 recent depth frame, at the cost of one frame of depth staleness. Audio synthesis remains excluded.

Table 4: Input-channel ablations. Shadow gain at 5 s (micro, per tier) and rollout shadow gain at 5 s for the full model and the no-visibility and no-flow variants.

variant	shadow gain @ 5 s (micro, static/moving/debris)	rollout shadow gain @ 5 s
full (occupancy+visibility+flow)	+0.306 / +0.363 / +0.315	+0.320
no visibility	+0.309 / +0.360 / +0.313	—
no flow	—	+0.381

Anticipating occupancy revealed in shadow (a falling-object *proxy*). We measure whether the model flags occupancy that materializes in unobserved space before it is revealed. Cell-level detection is 0.787 at a horizon-granular mean lead of 2.62 s. A component-level variant that scores each connected arrival region gives detection 0.752 and exposes what the cell-level false-alarm rate (0.021) hides: component-level precision is only 0.714. Three definitional caveats apply: events are counted per due prediction (one object aloft is scored at every issue step and horizon it is due, not once per object); “arrival” keys on unobserved-at-issue rather than newly-occupied cells, so occluded static geometry revealed by agent motion also counts and the metric pools all tiers; and precision is an unmatched majority-overlap over predicted components, sensitive to fragmentation (both scorings tabulated in Appendix C; qualitative panels in Appendix F). Lead time is bounded by the $\{1,3,5,10\}$ s horizon grid — an interface limit — and the ≥ 3 s target is not met. We report this metric as a diagnostic, not a strength.

Navigation. With the completed grid feeding the planner, the trained reactive policy achieves a per-episode collision rate of 0.367 (vs 0.60 untrained) and path straightness 0.773, but does not meet the ≤ 0.10 collision target. We present navigation as an honest system demonstration.

A generative (conditional-diffusion) variant of the world model is deferred to Appendix D.

5 Related work

Predicting the structure of unobserved space has been approached from three angles. Semantic scene completion, from SSCNet [Song et al., 2017] through MonoScene [Cao and de Charette, 2022] and its monocular successors VoxFormer [Li et al., 2023] and VisHall3D [Lu et al., 2025], infers dense voxel geometry — and even hallucinates geometry beyond the visible frontier — from a single image, but targets outdoor driving voxels evaluated purely by mIoU/IoU, without calibrated uncertainty in the occluded region. Occupancy world models such as OccWorld [Zheng et al., 2024], together with occupancy-forecasting benchmarks (Occ3D; Tian et al., 2023) and self-supervised raycasting forecasters [Khurana et al., 2023], instead predict the *temporal* evolution of multi-frame occupancy for autonomous vehicles. Closest in spirit, ProxMaP [Sharma et al., 2023] completes indoor proximal occupancy from a single view for navigation, yet emits only point estimates. ShadowWeave is distinct in delivering egocentric BEV amodal completion into explicitly masked occluded space from a single monocular-depth frame, with calibrated uncertainty inside the hidden region, and a completion-versus-forecasting decomposition showing the gain is spatial completion rather than dynamics.

6 Limitations

Our evaluation is in simulation; ground-truth occluded occupancy is unavailable on real footage without instrumentation, so no quantitative real-world claim is made. The perception front-end nonetheless accepts real monocular depth (Depth-Anything-V2), so sim-to-real transfer of the completion model is the natural next step — noting that evaluation currently consumes the simulator’s clean depth, so robustness of the completion to estimated-depth noise is untested. We compare against strong self-constructed baselines rather than prior methods, which target different input regimes (e.g., ProxMaP’s top-down RGB-D view); a head-to-head on a shared benchmark is future work. Our

results characterize a single 31M single-frame CNN: the evaluation protocol is architecture-agnostic, and the negative result on forecasting applies to this model class, not necessarily to larger or recurrent models. The model performs amodal *completion*, not multi-step *forecasting*: the pure-dynamics residual is at the noise floor, and we frame the contribution accordingly. Navigation does not meet its collision target and the audio interface is not user-evaluated; both are presented as system context. Falling-object lead time is bounded by the discrete horizon grid.

7 Conclusion

Calibrated amodal completion of occluded occupancy is achievable from a single monocular-depth frame, transfers to room geometries the model never saw, and can be delivered within a real-time perception-to-planning budget for an eyes-free interface. Equally important is *how* we measure it: an empty-credit-immune micro-gain, per-episode confidence intervals, and a decomposition that separates completion from forecasting keep the claim honest and expose exactly where the predictive power lies. We hope the protocol is reused: many occupancy-completion results would read differently under it.

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A Per-horizon shadow gains with confidence intervals

Table 5 reports the micro-averaged shadow-IoU gain over persistence at every horizon, with 95% per-episode bootstrap confidence intervals (10,000 resamples, paired model–persistence). Every interval is clear of zero. The static-tier gain is flat across horizons (persistent structure does not move); the debris-tier gain decays with horizon as more debris settles into persistent structure; the randomized-geometry run tracks the fixed-geometry static tier throughout.

Table 5: Micro shadow gain over persistence, all horizons, 95% bootstrap CIs over episodes.

tier	1 s	3 s	5 s	10 s
static	+0.307 [+0.288, +0.327]	+0.307 [+0.288, +0.327]	+0.306 [+0.288, +0.327]	+0.306 [+0.287, +0.327]
moving	+0.361 [+0.348, +0.375]	+0.364 [+0.351, +0.379]	+0.363 [+0.349, +0.378]	+0.361 [+0.347, +0.375]
debris	+0.343 [+0.315, +0.370]	+0.329 [+0.297, +0.361]	+0.315 [+0.281, +0.350]	+0.290 [+0.256, +0.327]
rand. geometry	+0.313 [+0.299, +0.328]	+0.312 [+0.298, +0.327]	+0.309 [+0.294, +0.324]	+0.303 [+0.287, +0.318]

B Heuristic baselines: observation-blind prior and dilation

The pose-marginal prior is the mean training-set occupancy grid in the egocentric frame, thresholded at whichever of eight thresholds (0.02–0.5) maximizes its micro shadow IoU — the most favorable possible reading. Table 6 shows that even so, the prior can reach high structure recall only by blanketing ~ 72 –79% of genuinely empty hidden space with false positives, capping its IoU at 0.08–0.10 versus the model’s 0.61–0.72. One caution: the prior’s IoU restricted to *dynamic* cells should never be quoted in isolation — at its best threshold the prior carpet-bombs the dynamic region too, which is meaningless at a 72% false-positive rate.

Table 6: Observation-blind prior at its best threshold (0.05) vs the model, @5 s.

tier	prior micro IoU	prior structure recall	prior FP on empty	model (IoU @ FP)
static	0.102	0.879	0.723	0.610 @ 0.034
moving	0.080	0.865	0.722	0.723 @ 0.015
debris	0.089	0.869	0.717	0.670 @ 0.014
rand. geometry	0.087	0.921	0.793	0.655 @ 0.021

The observation-*conditioned* counterpart — binary dilation of the observed occupancy by $r \in \{1, \dots, 6\}$ cells (one cell ≈ 10 cm), reported at the IoU-maximizing radius — probes whether local extrapolation of the observation suffices. It does not (Table 7): the heuristic peaks at $r=1$, essentially tied with or below plain persistence, because at matched false-positive rates it recalls only 0.42–0.56 of hidden structure versus the model’s 0.83–0.93, and every larger radius grows false positives faster than recall, monotonically lowering IoU.

Table 7: Observation-conditioned dilation heuristic at its best radius ($r=1$) vs persistence and the model, micro shadow IoU @5 s.

tier	dilation (best r)	dilation recall @ FP	persistence	model
static	0.305	0.416 @ 0.035	0.303	0.610
moving	0.346	0.557 @ 0.039	0.360	0.723
debris	0.342	0.544 @ 0.039	0.355	0.670

C Falling-object proxy: cell-level vs component-level

The component-level variant splits each frame’s arrival mask into connected components and scores each independently, which exposes what the cell-level false-alarm rate hides: of the distinct arrival components the model predicts, 29% are spurious. All the definitional caveats of §4.7 apply to both

Table 8: Cell-level vs connected-component scoring of the revealed-occupancy proxy.

metric	cell-level	component-level
detection rate	0.787	0.752
false-alarm rate	0.021	0.286
precision	—	0.714
mean lead time (horizon-granular)	2.62 s	2.63 s

columns: events are per due prediction rather than per physical object, “arrival” keys on unobserved-at-issue rather than newly-occupied cells (so occluded static geometry revealed by agent motion counts, across all tiers), and precision is an unmatched majority-overlap over predicted components, sensitive to fragmentation.

D Generative variant (ongoing work)

We additionally trained a conditional-diffusion (DDPM) world model as an alternative to the deterministic U-Net, motivated by the fact that BCE is mean-seeking and hedges into a grey smear exactly where the future is multimodal — the shadow — whereas a generative model can sample coherent alternatives and expose its disagreement as uncertainty concentrated in the occluded region. A full calibrated evaluation (in-shadow sample-diversity ratio and calibration, the fair comparison given that BCE and DDPM optimize different objectives) is ongoing: iterated DDIM sampling exceeds our batch-eval budget, and we leave a like-for-like generative comparison to future work. The deterministic model’s calibrated per-cell probabilities already carry the uncertainty narrative of this paper.

E Reproducibility

Scene generation for the fixed-geometry environment is pinned by a golden-hash regression test (byte-identical XML for the first 50 seeds), and a full 90-episode rollout re-run reproduces every published accuracy metric bit-identically (wall-clock latency, which is hardware-dependent, is the one exception). All full-model numbers use a single early-stopped checkpoint selected on validation IoU; train, validation, and evaluation episodes use disjoint seed ranges. Code, configuration, seeds, the evaluation summaries behind every table, and a frozen dependency lock accompany the paper; datasets are regenerable from seeds with a single pipeline script.

F Qualitative examples

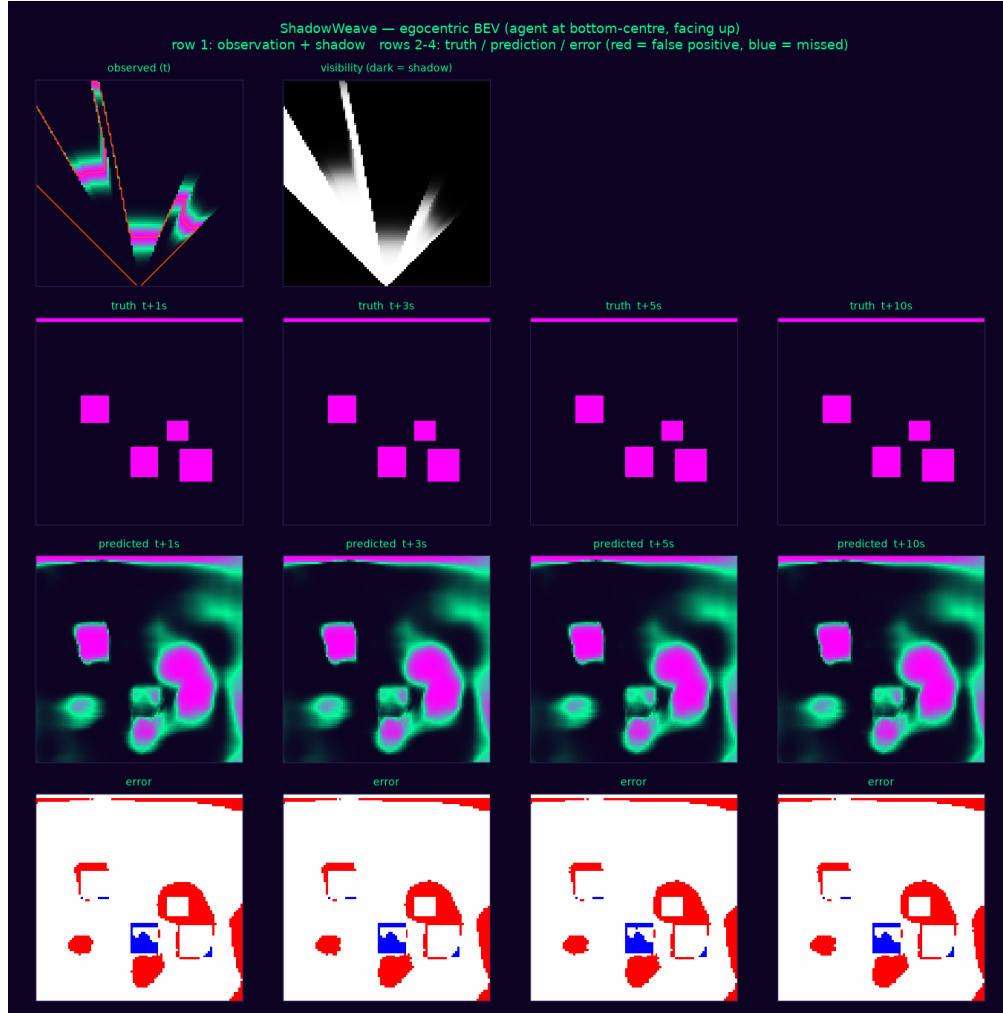


Figure 5: Qualitative completion panel (static tier). Row 1: observed occupancy at issue time and the visibility grid (dark = shadow). Row 2: ground-truth occupancy at $t+1/3/5/10$ s, rendered in the issue pose. Row 3: model predictions — the forecasts fill the shadow cones with the completed structure. Row 4: error maps (red = false positive, blue = missed).